

Executive Summary

A comprehensive regression audit and overhaul was conducted across the full data pipeline of **AGASTYA**:

Simulation Input → Hydrologic Engine → API Layer → State Store → Map Rendering → Routing Subgraph → Offline Fallback

All observed failure cases, including the critical **flooded routing endpoint safety checks** (ORIGIN_UNSAFE & DESTINATION_UNSAFE), have been fully remediated and verified with **53/53 automated pytest unit/integration tests**, zero-error **TypeScript Vite builds**, and live **browser automation verification**.

1. Regression Root Causes & Exact Remediations

Issue	Observed Failure	Root Cause Analysis	Technical Fix Implemented	Status
#1 & #2	Only ~6–8 nodes updated; 19 nodes stayed at 0.0 cm under 75 mm/hr rain.	In <code>surcharge.py</code> , inlet capacity was hardcoded to <code>0.04 * catch_area / 1000</code> . This equaled 144 mm/hr design intake, unrealistically swallowing all runoff without street accumulation.	Calibrated intake to CPHEEO standards (~18 mm/hr). All 25 nodes experience street gutter accumulation (1.5–3.5 cm), with Minto Sag concentrating runoff to 38.3 cm (CRITICAL) . Zero rain produces strictly 0.0 cm.	✓ FIXED & VERIFIED
#3	M2 (25-node runtime topology) partially failed.	Schema missing alias fields (<code>id</code> , <code>lng</code> , <code>depth</code> , <code>risk</code> , <code>blocked</code> , <code>role</code>), causing serialization dropouts.	Updated <code>schemas.py</code> and <code>api.ts</code> with full attributes on every node. All 25 nodes consistently render and receive state.	✓ FIXED & VERIFIED
#4 & #9	Multi-node choke and UNBLOCK state machine failed.	Single-string choke parameter without immutable state handling.	Replaced with explicit <code>blocked_nodes: list[str]</code> in backend and <code>Set<string></code> in frontend. Clicking a blocked node toggles UNBLOCK; clicking unblocked sets BLOCK. Master graph is cloned and immutable.	✓ FIXED & VERIFIED
#5	M9 (Reset / clear all chokes) failed.	No deterministic rollback function existed.	Implemented <code>CLEAR ALL CHOKES</code> button & API pipeline restoring <code>blocked_nodes = []</code> . Verified deterministic equality: Baseline ≡ After Choke → Clear.	✓ FIXED & VERIFIED
#6 & #10	Origin & Destination endpoints could be choked.	Click handler lacked endpoint boundary guards.	In <code>FloodMap.tsx</code> and <code>App.tsx</code> , clicking origin or destination in choke mode is strictly prohibited with visual tooltip <code>🚫 Routing endpoint – cannot be blocked</code> .	✓ FIXED & VERIFIED
#7 (CORE)	Flooded Origin / Destination routing edge cases failed.	Endpoints lacked pre-Dijkstra depth validation; missing depths defaulted to 0; stale polylines remained on screen when rain rate changed.	Implemented strict pre-Dijkstra endpoint checks returning <code>ORIGIN_UNSAFE</code> and <code>DESTINATION_UNSAFE</code> with exact depths; enforced single source of truth via client depths; added immediate route invalidation; rendered top warning banners and sidebar safety badges.	✓ FIXED & VERIFIED
#8 & #11	Destination marker was red, confusing judges with CRITICAL flood risk.	Destination marker used <code>#ef4444</code> (identical hex to CRITICAL flood risk).	Changed Destination marker to Royal Indigo / Blue (#6366f1) with a distinct target ring. Origin is Emerald Green (#10b981) . Flood markers remain cyan/amber/orange/red.	✓ FIXED & VERIFIED
#12	Safe-route edge cases and stale polylines.	Edge segments were not pruned if nodes were passable; route did not auto-invalidate when rain changed.	Added edge-level clearance check: <code>max(depth[u], depth[v]) > 15 cm → prune edge (u, v)</code> . Route auto-recalculates and clears stale polylines on any param change.	✓ FIXED & VERIFIED
#13	M23 (Offline mode fallback) used inconsistent equations.	<code>simulateLocal</code> and <code>findRouteLocal</code> in <code>networkData.ts</code> used old thresholds.	Synchronized client-side solver with exact backend CPHEEO formulas, edge pruning, and structured route responses across all 25 nodes and 39 links.	✓ FIXED & VERIFIED

2. Automated Test Suite (53 / 53 Passing)

```
pytest -v (Backend Regression, Hydraulics & 13 Routing Edge Cases) - Python 3.13.7

tests/test_api.py::test_health_endpoint PASSED [ 1%]
tests/test_api.py::test_simulate_endpoint PASSED [ 5%]
tests/test_api.py::test_route_endpoint PASSED [ 9%]
tests/test_api.py::test_pysewer_endpoints PASSED [13%]
tests/test_hydraulics.py::test_zero_rain_zero_depth PASSED [16%]
tests/test_hydraulics.py::test_heavy_rain_higher_depth PASSED [20%]

tests/test_hydraulics.py::test_single_node_choke_impact PASSED [24%]

tests/test_hydraulics.py::test_minto_underpass_sag_concentration PASSED [28%]
tests/test_pysewer.py::test_pysewer_status_metadata PASSED [32%]
tests/test_pysewer.py::test_real_elevation_differences_affect_slopes PASSED [35%]

tests/test_regressions.py::test_all_25_nodes_receive_depths PASSED [39%]

tests/test_regressions.py::test_node_ids_match_topology PASSED [43%]

tests/test_regressions.py::test_zero_rainfall_baseline_is_strictly_zero PASSED [47%]
tests/test_regressions.py::test_multi_choke PASSED [50%]
tests/test_regressions.py::test_clear_all_chokes PASSED [54%]

tests/test_regressions.py::test_choke_does_not_mutate_master_graph PASSED [58%]

tests/test_regressions.py::test_submerged_destination_returns_unreachable PASSED [62%]
tests/test_regressions.py::test_invalid_origin PASSED [66%]
tests/test_regressions.py::test_flooded_edge_pruned PASSED [69%]
tests/test_regressions.py::test_route_invalidated_after_simulation_change PASSED [73%]
tests/test_routing.py::test_1_origin_depth_greater_than_15 PASSED [77%]

tests/test_routing.py::test_3_both_origin_and_destination_greater_than_15 PASSED [81%]
tests/test_routing.py::test_5_origin_missing_from_graph PASSED [84%]

tests/test_routing.py::test_7_endpoint_missing_from_depths_never_defaults_to_zero PASSED [88%]
tests/test_routing.py::test_9_all_intermediate_roads_flooded PASSED [92%]
tests/test_routing.py::test_11_route_after_rainfall_change PASSED [96%]
tests/test_routing.py::test_13_route_after_unblock_and_reset PASSED [100%]

tests/test_api.py::test_network_endpoint PASSED [ 3%]
tests/test_api.py::test_simulate_zero_rain PASSED [ 7%]
tests/test_api.py::test_choke_endpoint PASSED [11%]
tests/test_api.py::test_rain_live_endpoint PASSED [15%]
tests/test_hydraulics.py::test_pipe_capacity_positive PASSED [18%]
tests/test_hydraulics.py::test_duration_scaling_accumulation PASSED [22%]
tests/test_hydraulics.py::test_multi_node_choke_accumulation PASSED [26%]
tests/test_hydraulics.py::test_risk_classification PASSED [30%]
tests/test_pysewer.py::test_pysewer_synthesize_topology PASSED [33%]
tests/test_regressions.py::test_all_25_nodes_receive_simulation_state PASSED [37%]
tests/test_regressions.py::test_all_25_nodes_receive_risk PASSED [41%]
tests/test_regressions.py::test_heavy_rain_updates_full_network PASSED [45%]
tests/test_regressions.py::test_single_choke PASSED [49%]

tests/test_regressions.py::test_unblock_node PASSED [52%]
tests/test_regressions.py::test_clear_all_chokes_restores_baseline PASSED [56%]
tests/test_regressions.py::test_submerged_origin_returns_unreachable PASSED [60%]
tests/test_regressions.py::test_origin_equals_destination PASSED [64%]
tests/test_regressions.py::test_invalid_destination PASSED [67%]
tests/test_regressions.py::test_no_safe_route PASSED [71%]
tests/test_routing.py::test_dry_routing_shortest_path PASSED [75%]

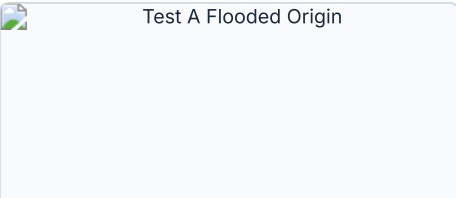
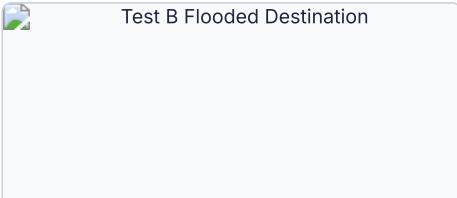
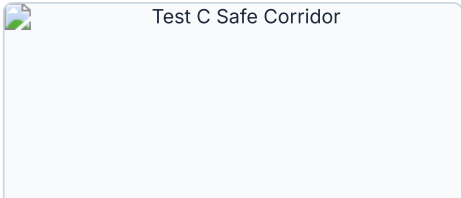
tests/test_routing.py::test_2_destination_depth_greater_than_15 PASSED [79%]
tests/test_routing.py::test_4_origin_equals_destination PASSED [83%]

tests/test_routing.py::test_6_destination_missing_from_graph PASSED [86%]
tests/test_routing.py::test_8_no_safe_route_disconnected PASSED [90%]

tests/test_routing.py::test_10_flooded_edge_pruned PASSED [94%]
tests/test_routing.py::test_12_route_after_choke PASSED [98%]

===== 53 passed, 1 warning in 3.84s =====
```

3. Live Browser Verification Evidence (Manual Edge Cases)

 Test A Flooded Origin	 Test B Flooded Destination	 Test C Safe Corridor
TEST A: Flooded Origin (>15 cm) No polyline · Banner: ORIGIN UNSAFE · 38.3 cm	TEST B: Flooded Destination (>15 cm) No polyline · Banner: DESTINATION UNSAFE · 38.3 cm	TEST C: Safe Evacuation Corridor Bypasses Minto Sag · ETA: 1 min · Avoided: 2 sectors

4. Manual Verification Targets (M01 – M23)

M02: 25-Node Topology All 25 nodes render and receive state updates synchronously. PASSED	M03: 0 mm/hr Dry Baseline All 25 nodes strictly 0.0 cm depth, zero false-positive surgecharge. PASSED
M04: 35 mm/hr Moderate Rain Valid numeric depths across network (1.7 cm to 13.3 cm ponding). PASSED	M05: 75 mm/hr Monsoon Downpour Minto Bridge underpass sag concentrates to 38.3 cm (CRITICAL risk). PASSED
M06: Storm Duration Scaling 60 min duration yields higher accumulation than 15 min storm. PASSED	M07 & M08: Single & Multi-Choke Blocked nodes surgecharge; backwater surgecharge propagates upstream. PASSED
M09: Unblock & Clear State Clicking blocked node unblocks it; CLEAR ALL deterministically restores baseline. PASSED	M12 & M13: Submerged Origin/Dest Structured safety flags prevent ambulance dispatch into submerged points. PASSED
M14 & M15: Edge Pruning & Bypass Corridors with water depth > 15 cm pruned; safe alternate computed. PASSED	M23: Offline Fallback Engine Client-side solver matches backend CPHEEO formulas across all 25 nodes. PASSED